

SCHOOL OF ENGINEERING, TECHNOLOGY & DESIGN

ASSIGNMENT GUIDELINES

TITLE OF MODULE: Big data database development	MODULE COMPONENT: 30% of Module
MODULE TEAM: Dr. Amina Souag Mrs Merlin Kasirajan	ASSIGNMENT CONTACT: Dr. Amina Souag
ASSIGNMENT DEADLINE : 15/04/2024	EXPECTED FEEDBACK DATE: 06/05/2023. Location of Feedback: Via Turnitin on Blackboard
ASSESSMENT TYPE Written report	
WHERE TO SUBMIT: TURNITIN SUBMISSION WHAT TO SUBMIT: A report with high professional standard	
TITLE(S) OF ASSIGNMENT: Assignment 1 – Database design	

Overview

A comity gathering the local food organizations across England set a mission to create an environment in which all citizens and visitors can access and know about good food that is sustainably grown, produced, processed, distributed, sold and served within each county. To achieve this, the comity would like to develop an online database where users can access information about local food and drink producers, suppliers, products, local markets, etc. and that per each county.

As a development team within Data Tech Analytics, you have been assigned the task of investigating whether a NoSQL document model and database would be the best solution. The goal is to create a scalable big data document database that would be used to store details of the various food and drink retailers and their products as well as details of the suppliers, local markets, etc. The database will be later used to produce analytical insights into the food and drinks industry in England.

Assignment Instructions

This assignment assesses your ability to apply the techniques that have been delivered through this module.

There are four key aspects that you are expected to deliver on completion of this assignment:

1. **Requirements analysis:** Fully documented requirements analysis.
2. **Conceptual model:** A fully formed and annotated Entity/Relation (ER) conceptual model.
3. **Denormalization:** Explain how you can 'migrate' from the obtained Entity/Relation model to a document model and provide the resulting Json document model.
4. **Queries:** Show how you can create and query your document database with two MongoDB (MQL) queries.
5. **Final report:** A written report outlining your approach and methodology. Your report should provide at the end a conclusion on why a document database would be a good or a bad solution for this project.

The initial Entity/Relation (ER) model has been provided in Appendix B. Please note that this model is primarily and far from being complete.

Note: It is recommended that you review the full set of learning outcomes and the rubric to ensure that all of the required information is included in your assignment submission.

Database Requirements Gathering

Your first task is to capture and fully understand the requirements for the new database.

It is recommended that you create and provide a succinct list of requirements that includes:

1. Requirement ID.
2. Requirement Name.
3. Requirement Description.

Once you have gathered your requirements, you will need to use them to design a final Entity/Relation data model. Your final ER will then be used to produce the document model using denormalization techniques.

The requirements provided by the comity representative in this assignment brief are **not** complete.

Once you have analyzed the provided requirements, you will need to email a business representative for the comity to finalize the full set of requirements.

The initial requirement descriptions provided are as follows:

- Users: users are required to sign up for an account. They will be able to create a username; registration will require an email.

- Products: each food and drink retailer sell different food and drink products. We need to record some core information about the products as well as the quantity of products sold per retailer.
- Food and drink retailer: We need to store the food and drink business name and core information such as the location, opening hours, etc..
- Food and drink Supplier: We need to record core information about the food and drink suppliers as well such as name of the business, turnover, number of stores and their locations, number of employees, products they supply ...
- Local market: We want to provide to the user the name of the local markets, products available, location of the market, the market's day, etc..

As outlined above, part of your project is to gather requirements from the representative for the businesses that hired you. You will need to email Tracy Bright in a professional manner to obtain the remaining requirements for the project. Tracy sits in the Operations Team within Data Tech Analytics and is keen to keep her clients and managers happy. Her contact details are as follows:

Tracy Bright
 Operation Manager
tbright.datatech@gmail.com

You will need to make a record of all communications with Tracy and submit them as part of your final report submission. It is down to you to determine what the client needs, and you will need to do this through consistent communication with their representative.

Appendix C contains some useful FAQs about contacting the representative. Please note that the module lead will not give details of the missing requirements. You need to contact the representative using clear and professional email communications.

During the last quarter of the tutorial sessions, your group will need to meet with the module lead to review your ER model and receive constructive feedback.

Finally, you will need to submit evidence of having used denormalization to produce your final document model.

Project Report and Documentation

You will be expected to provide a written report that documents your approach to gathering your requirements, the ER model, denormalization approach to produce the document model, a demonstration of how you can create and test your database with two queries.

One recommended report structure is as follows:

- Introduction.
- Project Requirements.
- ER model.

- Denormalization to produce the document model.
- Discussion on implementation strategies of the document database and testing with two queries.
- Appendix A: History of Communications

Please note: This is not indicative. You may decide to add, remove or change sections, as well as making use of additional appendices as you see fit.

Submission Guidelines

You will need to submit a report (Word document or PDF) via Turnitin to the assessment 1 area of the module on Blackboard.

CCCU Late Submission Guidelines

In the case of late submission of work for a component of assessment (including dissertations, extended essays, and projects) without an approved extenuating circumstance, the work will be penalized.

The standard penalty to be applied by all courses will be 10 percent, for any late submission up to 2 working days (unless a significant system outage occurs). The deduction will reduce the mark to no less than the pass mark, where the work would otherwise have passed. Where the work would otherwise have failed no further late penalties will be applied. After 2 working days late a mark of 0 will be recorded and the work will be considered as a non-submission. These regulations will apply, unless:

EITHER

an Additional Course Regulation has been approved by Academic Board;

OR

the work is marked on a pass/fail basis and it is not possible to give a numerical mark.

The submission deadline is 2pm on the date due. The application of penalty starts after 2:01pm on the day of the published deadline. For example, for late assessments, a penalty of 10 percent will be applied from 2:01pm on the day of the published deadline until 2pm on the second working day. The calculation of “working days” is Monday to Friday.

The 10 percent penalty will be 10 percent of the full one hundred percent and not 10 per cent of the mark awarded to the student. If, therefore, the student has scored 50%, the student will lose 10 percent and receive 40% where the work is submitted in the late period.

Please read the latest marking procedures: <https://www.canterbury.ac.uk/quality-and-standards-office/docs/marketing-procedures/Marking-Procedures-2324.pdf>

Mark Scheme/Rubric

Please see page 7 for a detailed rubric.

Formatting of Submission

You are expected to produce your report to a high professional standard. Any screenshots, tables, figures, charts, illustrations, etc. will not contribute towards the word count.

It is very important that you make sure that your figures are clear and visible to the reader.

Your work must be adequately referenced throughout, using Harvard referencing style. Pears & Shields (2016) give a complete guide to Harvard referencing. Guidelines on using the Harvard Referencing style are available at:

<https://www.canterbury.ac.uk/library/docs/harvard.pdf>

<https://www.canterbury.ac.uk/students/docs/study-skills/resource-1-Harvard-Referencing-Guide.pdf>

The report must be submitted using the dedicated Blackboard grade center submission bucket on or before the submission deadline.

Resources You May Find Useful

The majority of the resources you may need are accessible via the Big Data database development module on Blackboard at <http://learn.canterbury.ac.uk>.

Suggested tool to draw the ER diagram: Draw.io

Learning Outcomes Assessed (Fully or Partially)

1. Demonstrate comprehensive knowledge and understanding of Data Modelling for Big Data.
2. Demonstrate critical awareness of current problems and/or new insights in the field of Big Data.
3. Critically apply knowledge of core principles, concepts, and methods to design a database that responds to an industry relevant challenge.
4. Work effectively independently and as part of a team to develop inter-connected and complex database solutions.
5. Critically evaluate their solution from commercial and end user perspective with respect to usability and utility.

GRADUATE / EMPLOYABILITY SKILLS GAINED:

This assessment is an opportunity to develop an understanding of the design of a big data database solution and document it.

This assessment has been designed to exercise your ability to work on your own and the ability of researching and analyzing.

CCCU GRADUATE ATTRIBUTES

Adaptable:

- Resilient and helps to bring about positive transformation in the face of continuous and rapid change.
- Able to apply their knowledge and skills to thrive within a globalised society.

Digitally Literate:

- Engages critically and creatively with academic and professional communications through a range of media.
- Participates in digital networks for learning and research.
- Manages their digital reputation and online identity.
- Confident in the use of digital devices, applications and services.

Effective Communicator

- Understands and expresses their ideas and thoughts with confidence using a variety of media.
- Develops rapport, respects diversity, has passion and can communicate with self-assurance.

Informed

- Explores concepts and issues related to their academic programme of study and beyond, that have both local and international significance.
- Aware of issues relating to social justice, ethics and wellbeing.
- Understands the concepts of environmental stewardship.

Innovative

- Independent critical thinker with an intellectual curiosity.
- Exercises initiative in applying thinking styles critically and creatively.
- Able to develop and defend their own ideas with the capacity to challenge assumptions where necessary

Professional

- Possesses integrity and is able to instil confidence in others.
- Lifelong learner; managing their own self development.
- Recognises and approaches complex problems by making reasoned, ethical decisions.
- Self-Aware
- Self-reflective and empathetic; being able to respect and appreciate

PROGRAMMES OF STUDY:

Msc Data intelligence

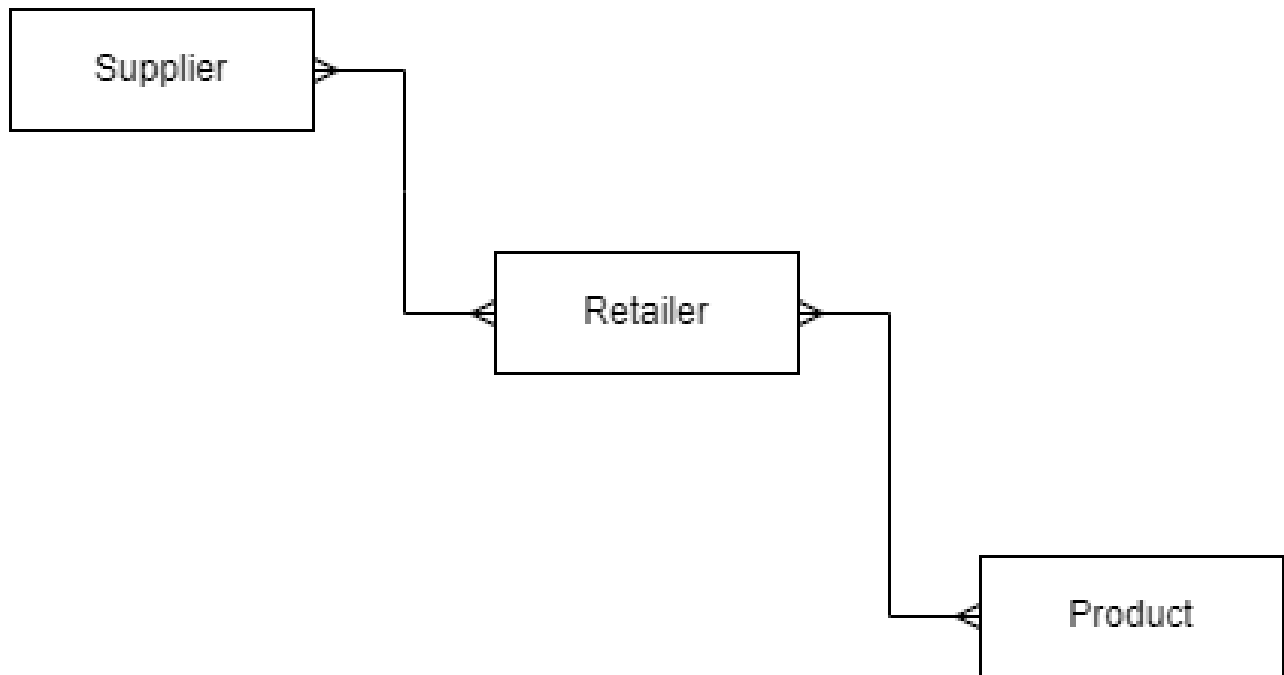
Appendix A - Grading rubric

Criterion	Fail 0-19	Borderline Fail 40-49	Pass 50-59	Merit 60-69	Distinction 70 -100
A. CONTENT, KNOWLEDGE & UNDERSTANDING					
Requirements analysis	No requirements analysis provided.	Requirements analysis is weak, descriptive and unstructured.	A few requirements were identified but they are limited in covering a good spectrum of the project.	Fully documented requirements analysis supported by evidence of exchange with the operation manager. Each requirement has an ID, is well expressed, and described.	Fully documented requirements analysis supported by evidence of exchange with the operation manager, enhanced with search of relevant documentation online. Each requirement has an ID, is well expressed, and described.
Conceptual (ER) model	The model fails to show details of entities and relations. There is no attempt to extend the provided model. Missing model.	The model extends the initial Entity-Relation model provided but further extension of additional entities is minimal. No relation between the requirements analysis and the ER model.	Able to translate most identified requirements into the ER model, able to identify most entities and their attributes as well as most of relations and cardinalities between entities.	Able to translate the identified requirements into the ER model, able to identify all entities and their attributes as well as most of relations and cardinalities between entities.	A fully formed ER model that fulfills all the requirements identified. Able to identify all entities and their attributes as well as all of relations and cardinalities between entities. All entities are appropriately named with correct relationships represented between them. Attributes are sensibly named, with suitable data types, defined nullability and any potential default values. The reader is left with a clear sense of what was done and why. Any limitations or weaknesses are explained.
Denormalization and of production document model	The documents are not adequately developed or missing. No use of denormalization to produce the document model	Some of the documents are only just adequately identified.	Able to identify some documents in the collection and justify choices of nesting.	Able to identify all documents in the collection and justify choices of nesting.	A fully formed collection of documents. Denormalization is adequately used to explain nesting choices. Able to identify all the documents, as

					well as their content properly structured. Any limitations or weaknesses are explained.
Queries	There is a lack of understanding in the formation of MQL queries. MQL queries are missing.	The MQL queries may not be complete, or the formatted correctly, but they are readable and there is evidence of understanding.	The queries are formatted correctly if not complete. Additional data may be retrieved that may busy the queries. The data retrieved is correct, but there may be some duplications.	The script runs without any interference and ultimately retrieves the required information. There may be more attributes than necessary. The data retrieved is correct.	The queries run as expected and outputs results for the user in a readable and clear manner. Appropriate data is selected to provide the necessary information to answer the queries. The script runs on MongoDB.
Cognitive skills					
Referencing	Little or no acknowledgement of sources of information in text and/or reference list.	Sources of information acknowledged but integration between text and reference list is inconsistent. Attempts to follow a systematic approach, appropriate to the discipline do not always succeed.	Effective use of sources to support discussion and personal findings. Referencing follows the standards of the discipline clearly and accurately.	Very effective use of sources to support discussion and personal findings. Referencing follows the standards of the discipline very clearly and accurately.	Outstanding use of sources to support discussion and personal findings. Referencing follows the highest standards of the discipline.
Originality in application of knowledge and in problem solving	The knowledge and approaches applied were not relevant or appropriate to addressing the task.	Knowledge and problem solving were applied largely accurately, but a more original approach and deeper discussion were required for the assignment.	Effective application of knowledge and problem solving, but with some original features.	Demonstrates some creativity and innovation in the application of knowledge and/or practice and in problem solving.	Demonstrates outstanding originality, creativity, and innovation in the application of knowledge and/or practice and in problem solving. Has definite potential for publication/exhibition and ability for doctoral research.
Conclusion	The conclusion is not adequately developed or missing.	The conclusions are only just adequately developed.	Conclusions are drawn which are clearly derived with some originality	Conclusions show very good development and original insight and relate clearly and logically to evidence of the work done.	Outstandingly well-developed conclusions which show considerable originality. They contribute to the work being of publishable standard.

TRANSFERABLE SKILLS					
Clarity of expression (incl. accuracy, spelling, grammar, punctuation)	The writing style and structure are consistently disorganized and/or incoherent. The meaning is unclear. Discipline-specific vocabulary is not used appropriately.	The syntax and writing style are confusing for the reader and discipline-specific vocabulary may be misused or misrepresented. Frequent errors in spelling, punctuation, grammar, and numeracy.	Demonstrates fluent, effective, and confident writing with structured and coherent arguments. The writing style is adapted to the needs of the brief using discipline-specific vocabulary accurately to present complex ideas clearly. Spelling, punctuation, grammar, and numeracy are accurate.	Demonstrates very fluent, effective, and confident writing with well-structured and coherent arguments. The writing style is adapted effectively to the needs of the brief using discipline-specific vocabulary accurately to present complex ideas clearly. Spelling, punctuation, grammar, and numeracy are accurate.	Demonstrates very fluent, effective and confident writing with outstandingly well-structured, convincing and coherent arguments. The writing style is adapted sensitively to the needs of the brief using discipline-specific vocabulary accurately to present complex ideas clearly. Spelling, punctuation, grammar, and numeracy are accurate.

Appendix B: Initial Database ER-model



APPENDIX C: FAQ REPRESENTATIVE COMMUNICATION

We need information about <topic>, how do we know what questions to ask?

You need to open a dialogue with the representative to better understand where their expertise lies. Be sure to ask questions; if they do not know the answer, they will direct you to someone who can answer the question.

We have emailed Tracy and they have not responded, what should we do?

The client is communicating with you as well as working their everyday jobs; they are busy! As a result, your email may be lost amongst their vast inbox. It is therefore vital that you persist in your correspondents. If you are waiting for a reply, send another prompt to ask if they have read your last email. Try setting the importance of your mail; use High Importance. (Note, if you set all your mails as High Importance, they may filter them out. Ensure that you use this only for the most vital emails).

Do we have face-to-face meetings with “Tracy”?

There will be no face-to-face meetings, rather you will be required to communicate with the University Instructors who will act as an intermediary.